

CHAPTER 6

Distant Heat

6.1 FORGETTING TO BREATHE

As an observational astronomer, I get to travel to various supposedly exotic locations. People get envious when I tell them I'm going to Hawaii to observe, and they don't believe me when I tell them that, to me, Hawaii is a cold windswept mountaintop, working nights.

Of course it's not just the cold and the nights that are the problem. At 14,000 feet, the Mauna Kea Observatory on the Big Island is above a third of the atmosphere. Nothing about the human body works properly at this altitude unless you've done some serious acclimatization, and even then things can go wrong. Observers sleep, when they can, at the accommodation block at 10,000 feet, but this is never restful when you're spending twelve or more hours a night on the summit. And when you're on the summit things can get quite strange.

My first observing trip provided a useful demonstration of this. You expect to get out of breath easily, and the infamous staircase at the UK Infrared Telescope (UKIRT), from the control room down to the toilets, is a case in point. It's short enough to run up, but at 14,000 feet you should never try to do this unless you like hypoxia.

But what gets to you is the unexpected.

I am sitting at a workstation, examining some of my data, when I get the feeling I'm forgetting something important. I check around me to make sure the telescope's not stopped and that we're still taking data. No trouble there. I check the weather monitor to see if the clouds are rolling in. Nothing there either, and these problems seem rather too distant, too impersonal to be what's worrying me.

I'm forgetting something important, that I do all the time by instinct or reflex, but that isn't currently working.

Breathing.

Yes, that's it!

A few deep inhalations later and I'm feeling much better, and my CO₂ levels are back to the point where they can trigger the reflex again.

It's not just small problems like forgetting to breathe that make the mountain hazardous, though. You lose IQ points the higher you go, and this can affect you in very strange ways. It's the things you least expect that catch you out, even when it comes to the simple set-up procedures. For example, we managed to waste fifteen minutes one night working out how to align an instrument on the sky purely because we couldn't remember which way was east.

But observers have to face only a limited set of problems. The people building telescopes have to face unexpected problems throughout the day as they assemble some of the most complicated constructions on the planet. And altitude doesn't help.

Hence this conversation reputedly overheard at the construction site of an observatory that shall remain nameless:

Construction worker at the summit: We have a problem with this part. HQ at sea level: Oh yes? What is it?

CW: It's the wrong size.

HQ: The wrong size?

CW: Yes - I've cut it three times, and it's still too short!

HQ: Come down the mountain - now!

Fortunately the effects of the altitude aren't permanent. Once you're at sea level everything goes back to normal. In fact, it's a little better than normal after any time at the summit. You're suddenly reminded just how much air there is, how much oxygen there is in it, and that breathing should not be a major form of exercise.

6.2 GALAXIES STARS AND SO MUCH MORE

Everything we have looked at so far has been in our own Galaxy, the Milky Way, but this is just one of billions of galaxies in the observable universe. If we are to understand where we came from, how our own galaxy formed, and what its future is likely to be, then we have to look further afield, and study the properties of other galaxies.

Edwin Hubble was one of the first astronomers to seriously study galaxies, and to realise that they did not lie inside the Milky Way but were similar objects to it, much further away. There are two very different classes of galaxy, as defined by their shape: elliptical galaxies, and spiral galaxies. The Milky Way is a spiral galaxy. Hubble also devised a broader classification scheme which separates spiral and elliptical galaxies into separate types, and which separates spiral galaxies into those with and without a central bar-like feature. These are known as barred spirals. As with any taxonomic scheme there are, of course, exceptions, which Hubble labelled irregulars. [Figure 6.1](#) shows this classification scheme. The original idea was that elliptical galaxies evolved into spiral galaxies, so ellipticals are also known as 'early type' galaxies whilst spirals are known as 'late type'. This evolutionary scheme has since been found to be wrong - if anything the evolutionary processes work in the opposite direction, from spiral to elliptical - but, as with so many things in astronomy, we are now stuck with that nomenclature.

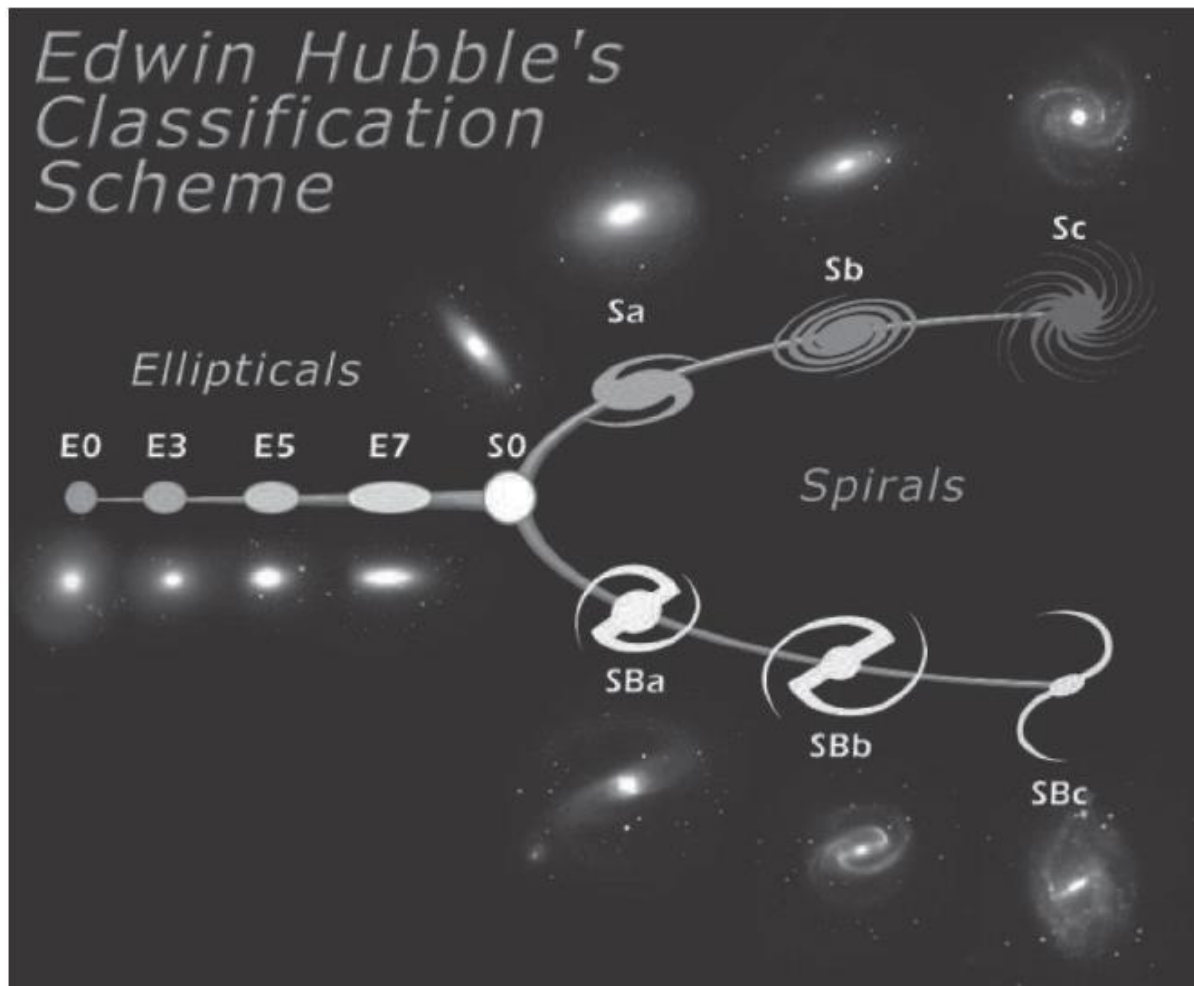


Figure 6.1 The Hubble 'tuning fork' classification scheme for galaxies.

(Courtesy of NASA.)

While the visible light that we see from stars in galaxies is the most obvious thing to our eyes, galaxies are much more than just agglomerations of stars. The majority of the mass of a galaxy is in fact made up of dark matter, the mysterious material whose gravitational interactions have defined the large scale structure of the universe. Each large galaxy sits inside a halo of dark matter and, in the outer regions of both spiral and elliptical galaxies, the motion of stars is controlled by the gravitational pull of dark matter, rather than that of other stars. Gas and dust can also lie between the stars, making up the galaxy's interstellar medium (ISM). The interstellar medium is the raw material from which new stars form, as we saw in the previous chapter. There is usually very little star formation in an elliptical galaxy, and they are largely made up of a stellar population that is evolving passively. This simply means that no new stars are being formed. Ellipticals are sometimes referred to as 'red and dead', their red colour coming from the lack of hot, and thus blue, young, high-mass stars, and dead referring to their lack of star formation. Spiral galaxies, in contrast, usually have large amounts of gas and dust, as well as active star formation, in their disks. Our own Milky Way Galaxy is a good example of this, with a star formation rate of about one solar mass of new stars being formed each year. Most of these new stars are, of course, low mass stars, given what we learnt about the stellar IMF in the last two chapters.

The gas in the interstellar medium of galaxies is similar to that found in the Milky Way. It is largely composed of hydrogen, as this is the most common element in the universe, and there will also be helium and a mixture of heavy elements that have been released into the interstellar medium by previous generations of stars. Hydrogen will be in one of three forms: molecular hydrogen, atomic hydrogen, and ionised hydrogen.

Ionised hydrogen can be detected in the optical and near-IR by its emission lines, while atomic hydrogen can be detected by its 21cm emission line in the radio. Molecular hydrogen, though, is rather more difficult to detect. It has a number of weak emission lines in the near infrared which can be excited in certain environments, but the more usual way to assess the total amount of molecular hydrogen is to look at a different molecule that acts as a tracer. The molecule usually used for this is carbon monoxide, CO, which has emission lines in the far-infrared and submillimetre. Observations at these wavelengths are thus needed to make a proper assessment of the amount and distribution of molecular gas in a galaxy.

Dust is also an important component of the interstellar medium in galaxies. Interstellar dust grains range in size from a few molecules to about 0.1 micron across. The smallest dust grains are effectively large molecules of polycyclic aromatic hydrocarbons (PAHs), which are a variety of carbon compounds made up of many interlinked hexagonal rings of carbon atoms. More exotic carbon molecules have been discovered in the interstellar medium as well, including fullerenes, the football-shaped molecules C_{60} and C_{70} named after Buckminster Fuller [63]¹. Large dust grains are made up of a carbon or silicate core onto which other molecules from the interstellar medium, including oxygen and nitrogen, have been adsorbed (see [Chapter 5](#)).

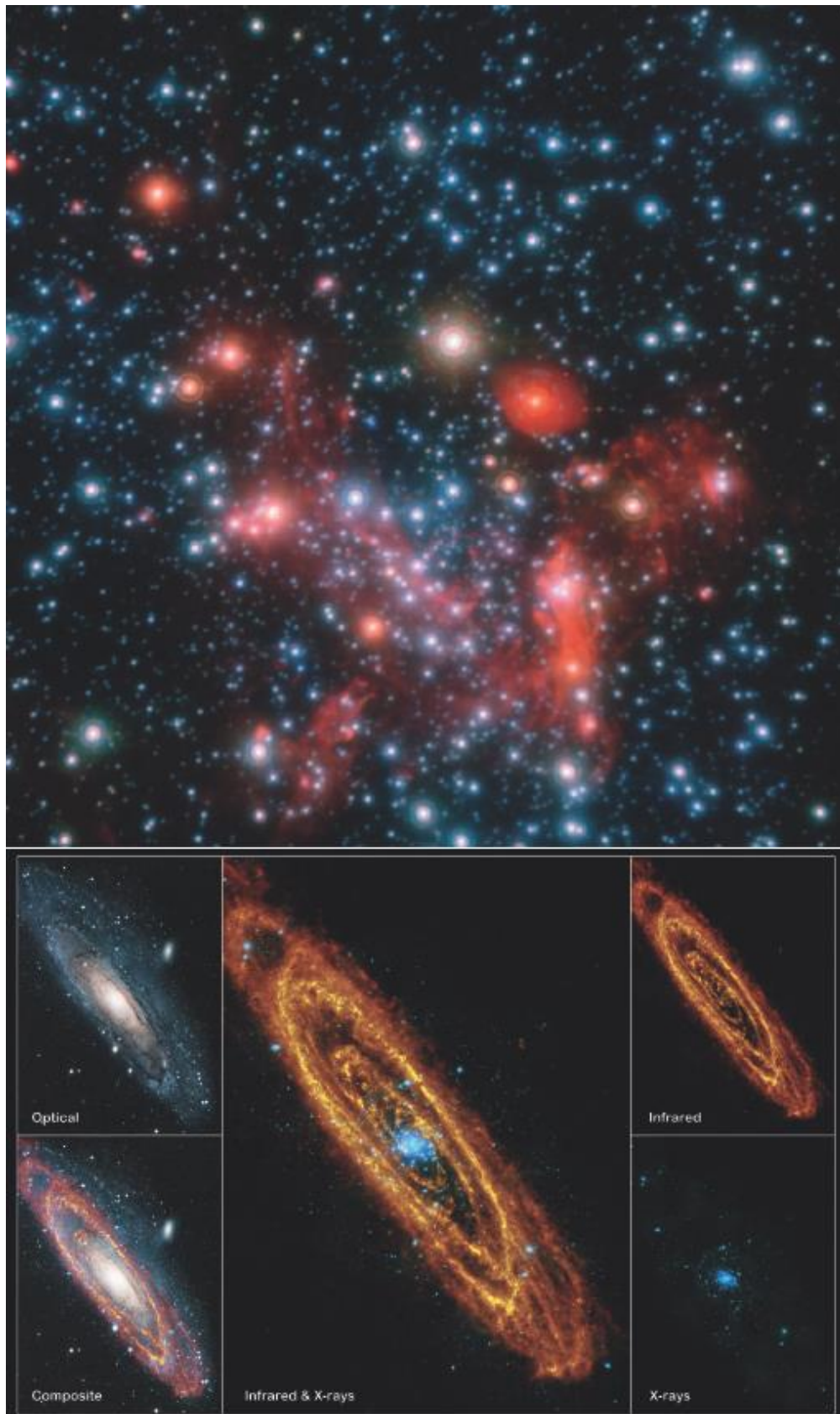


Figure 6.2 Top: A near-infrared image of stars at the centre of the Galaxy; Bottom: Our nearest neighbour, the Andromeda galaxy, seen in the optical, far-infrared and X-ray.

Top: The motion of these stars was monitored over 16 years and used to find the mass of the supermassive black hole at the centre of our Galaxy. (Courtesy of ESO). Bottom: The optical shows emission from stars, far-infrared shows emission from dust, X-ray shows the remnants left behind at the end of stellar evolution. (Courtesy of ESA.)

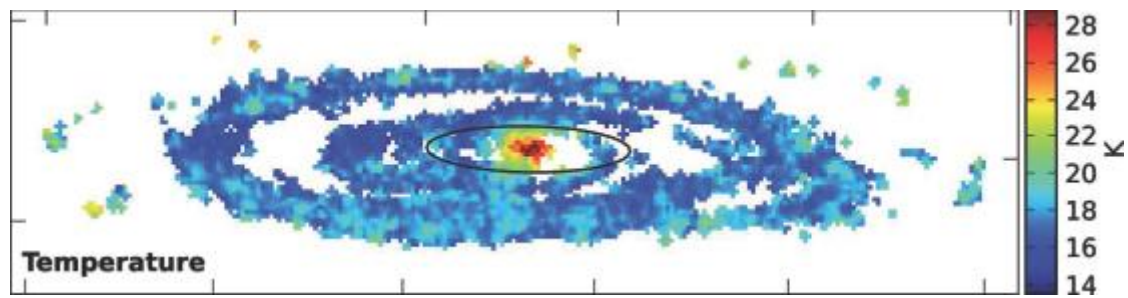


Figure 6.3 Dust temperature variation in the Andromeda galaxy.

(Courtesy of ESA.)

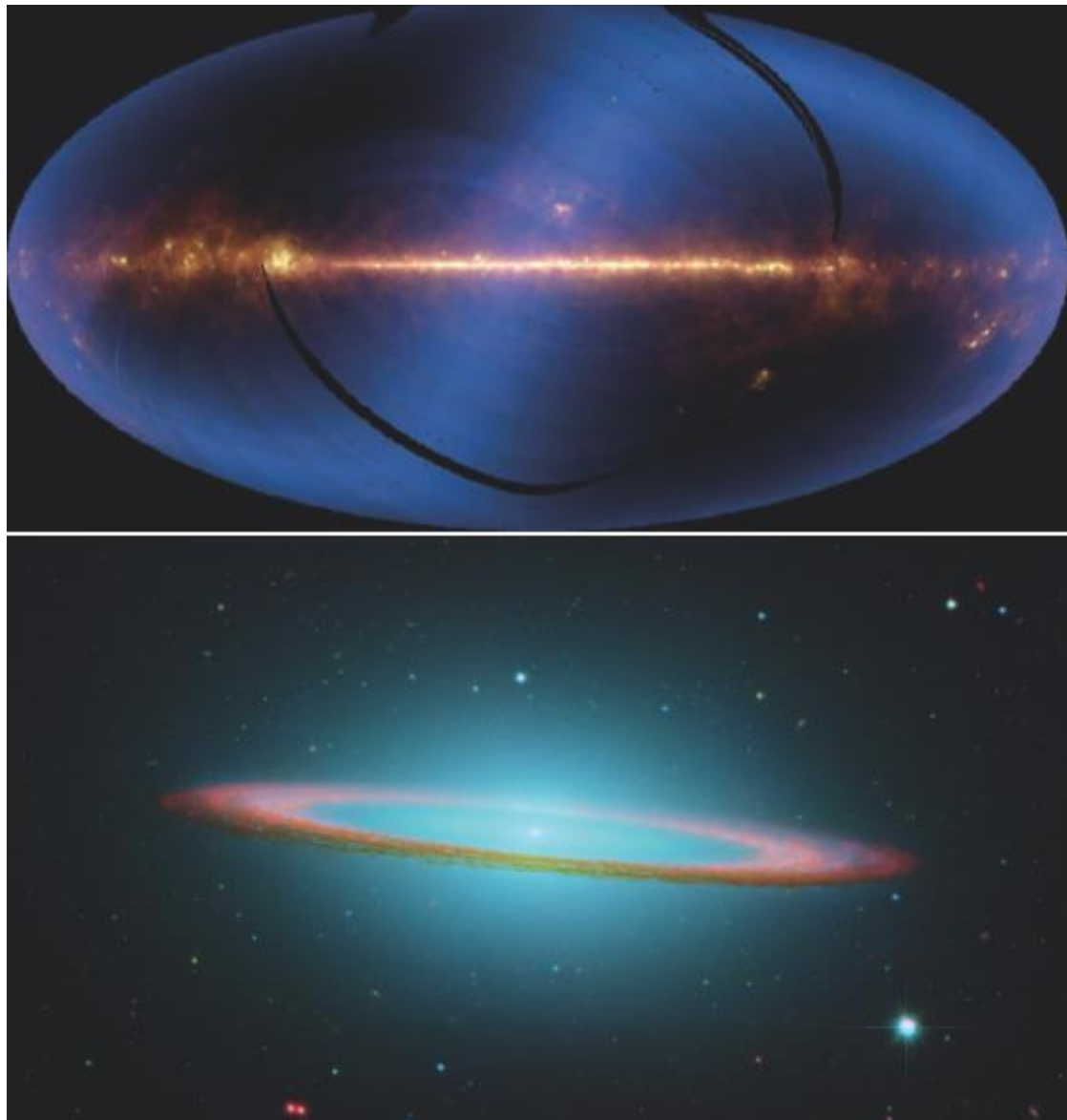


Figure 6.4 Top: The IRAS all-sky survey. Bottom: The Sombrero galaxy M104, in the optical and infrared.

Top: The whole sky from IRAS showing emission at 12 microns as blue, 60 microns as green, and 100 microns as red. The red band across the centre of the image is our own Galaxy. The blue streak across the centre to the edges is zodiacal dust in our own Solar System. (Courtesy of NASA.) Bottom: The Sombrero galaxy is shown in optical light (blue and green) from the HST, and infrared light (red) from Spitzer. While this galaxy is classified as a spiral, it has a very weak disk, but a very strong bulge. (Courtesy of NASA.)

There is one other constituent of most large galaxies that needs to be mentioned - the central supermassive black hole. Black holes are objects that have collapsed to such high densities that the velocity needed to escape from their gravitational attraction is greater than the speed of light. Normal, stellar-mass black holes are the end point of stellar evolution for some of the most massive stars. Such black holes are typically a few solar masses in size, though the remnants of supernova explosions of the most massive stars may be a few tens of solar masses. The black holes in the centres of galaxies, though, are many times this size. The black hole at the centre of our own Galaxy, for example, has a measured mass of about 4 million solar masses [64]. This was determined after a sixteen year programme monitoring the motions of stars at the very centre of our Galaxy that orbit this black hole. Optical telescopes can't do this since the dust and gas in the disk of the Milky Way obscures light from the Galaxy's centre. Near-infrared observations, which are less subject to this obscuration were used instead. [Figure 6.2](#) shows one of the infrared images used for this study, while a movie showing the movements of these stars over sixteen years can be found via this book's website.

While you may think the black hole at the centre of our Galaxy is huge, the black holes at the centres of other galaxies can be much larger. Black holes at the centre of giant elliptical galaxies have been found with masses up to ten billion solar masses [109], over a thousand times the mass of the black hole at the centre of the Milky Way. When matter falls onto a supermassive black hole, a large amount of energy is released. This may seem counter-intuitive, since light cannot escape from a black hole, but the energy released by in-falling matter is produced outside the event horizon, beyond which nothing can escape. This happens through the formation of a black hole accretion disk, a disk of matter spiralling ever faster as it falls towards the black hole. The process is not unlike the way water spirals around the plug hole in a bath, but much faster and much bigger. Friction between the different parts of this accretion disk heats the infalling matter to many millions of degrees, and this is how the energy of an accreting black hole is generated. The high temperature of the accretion disks in AGNs makes them prominent sources at X-ray wavelengths.

In most cases, galactic centre black holes in the local universe are quiescent: there isn't much matter falling onto them so they don't emit much energy. In a small number of cases, though, matter is being accreted at a much higher rate, producing what is referred to as an Active Galactic Nucleus (AGN). These objects have a wide range of properties which we will be discussing later in this chapter. While AGNs are rare locally, they were much more common at earlier stages of the universe. This is something we will look at in the next chapter.

6.3 GALAXIES ACROSS THE ELECTROMAGNETIC SPECTRUM

A galaxy's energy output as a function of wavelength, its spectral energy distribution (SED), as it is known, is dominated by different components of the galaxy at different wavelengths. The bluest stars are the most massive, with the highest power outputs, the hottest surfaces and, as we saw in [Chapter 4](#), the shortest lives. These are responsible for emission in galaxies at ultraviolet wavelengths, and at the blue end of the optical spectrum. Since these stars are short lived, only galaxies where there is star formation underway are bright in ultraviolet and blue light. This is the case for spiral galaxies like our own Milky Way. As can be seen from [Figure 6.5](#), which plots model SEDs for a number of different galaxy types, elliptical galaxies, lacking significant star formation, also lack the blue and ultraviolet emission that would come from young stars. At somewhat longer wavelengths, moving from the blue (short wavelength) end of the optical towards the red (longer wavelength) end, we reach wavelengths that correspond to lower temperature stellar surfaces, and thus cooler, longer-lived stars. The differences between elliptical galaxies and other types thus get smaller with increasing wavelength until in the near-infrared, where the emission is dominated by M-class stars, the SEDs are almost the same. Observations in the near-infrared can be used to determine the size of the underlying old population of stars in galaxies, with little contamination from the hot and bright, but short-lived, higher mass stellar population.

Beyond the near-infrared, we begin to see emission from material in the interstellar medium of galaxies, including gas, dust and PAHs. This material is not luminous in and of itself - unlike stars, it is not responsible for any energy generation. Instead, this material, in a variety of ways, absorbs light from stars and re-emits that energy at wavelengths different to those at which it was received. This is exactly what happens to sunlight when it reaches the earth. Rocks, sand or sunbathers on a sunny beach are drenched in light arriving from the Sun at optical wavelengths, peaking at roughly 500 nm in wavelength (0.5 microns). They absorb the energy in that light and warm up a little. As discussed in [Chapter 1](#), any object at a temperature above absolute zero emits electromagnetic radiation with a black body spectrum peaking at a wavelength given by its temperature. For the rocks, sand and sunbathers on a beach, that temperature will be about 300 K, equivalent to 20 degrees Celsius. A black body spectrum for an object at 300 K peaks at a wavelength of about 10 microns. This is well beyond the capabilities of our eyes to detect, which is why we can't see everything around us glowing, but it can be picked up by thermal imaging cameras, as seen in [Chapter 1](#). By absorbing light from the Sun and warming up a bit, our beach and sunbathers have, in effect, reprocessed the light of the Sun into mid-infrared radiation. In just the same way, the constituents of the interstellar medium absorb light from stars at ultraviolet, optical and near-infrared wavelengths, and reprocess it to longer wavelengths.

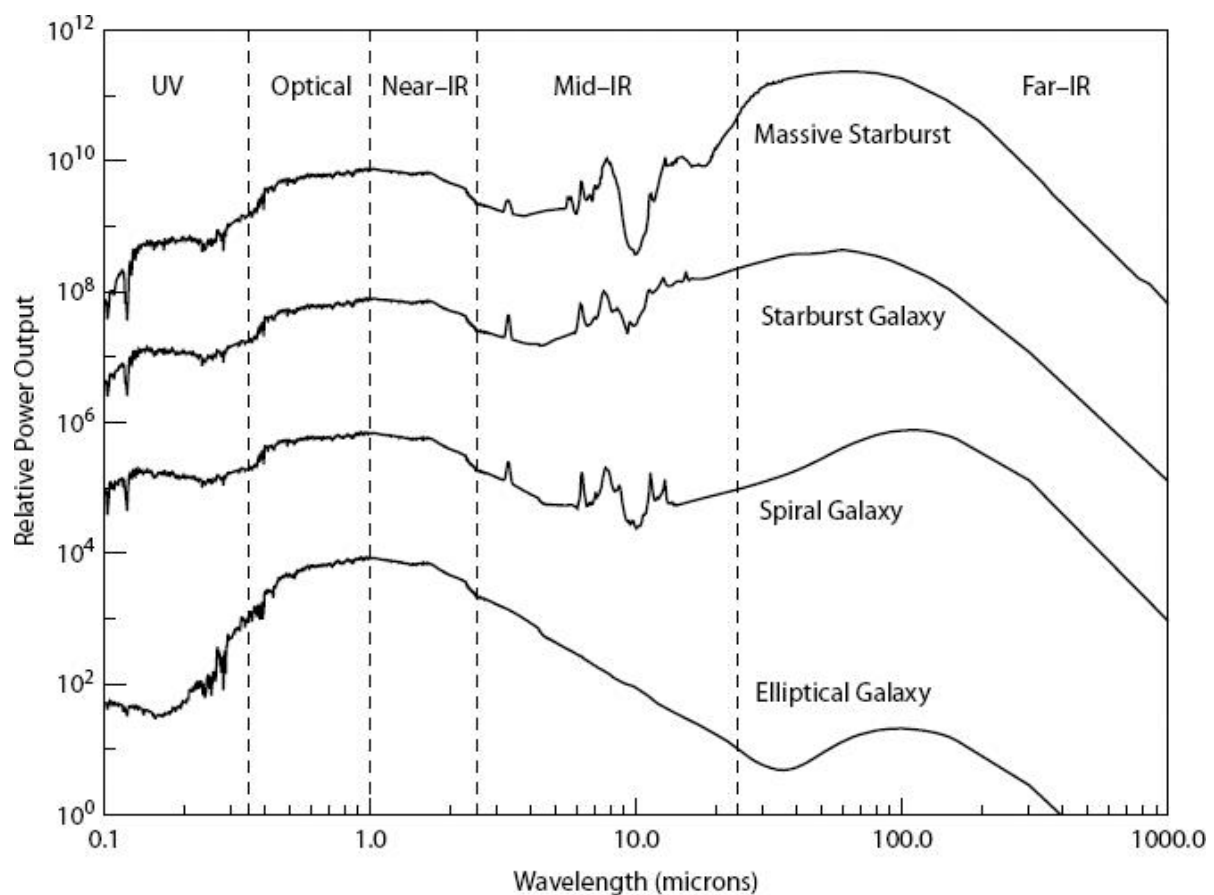


Figure 6.5 Model spectral energy distributions of several different types of galaxy.

The four different SEDs are shifted vertically so they can be compared. (Data for the model SEDs shown here are from [[135](#)]).

At the longest wavelengths, in the far-infrared, we see emission from dust at temperatures of 30 to 50 K, with emission peaking around 100 microns in wavelength. The dust responsible for this emission is made up of particles of size about 0.1 to 1 micron across. These particles may sit in the general interstellar medium between stars, or might be associated with denser regions of gas and dust responsible for star formation. The temperature of the dust will be somewhat warmer if it is

closer to powerful sources of heat, such as young stars or the galactic centre black hole, and cooler if it is far from such intense heat sources and is merely warmed by light from the general population of stars in a galaxy. The total mass of cool dust in a spiral galaxy can be quite large, up to a hundred million times the mass of the Sun, but an accurate measurement of this dust mass depends on an accurate measurement of the dust temperature, and on the balance between warmer components, close to star formation regions, and cooler components.

The mid-infrared spectral energy distribution of galaxies, as can be seen from [Figure 6.5](#), can be rather more complicated than the far-infrared. This is because several different components of the interstellar medium contribute to the shape of a galaxy's SED in the mid-infrared. First of all, there are the emission lines of different widths that can be seen as spikes upwards in [Figure 6.5](#). These are produced by PAHs - large carbon-based molecules floating freely in the interstellar medium. Various motions of these molecules produce strong broad lines centred on wavelengths of 3.3, 6.2, 7.7, 8.6, 11.2, 12.7 and 16.4 microns, as well as some other weaker features in the mid-infrared. PAH features are usually associated with regions of star formation, and can be used to investigate the star formation underway in galaxies.

As well as the PAH emission features, there is also continuum emission in the mid-infrared, corresponding to temperatures of a few hundred to ~ 1000 K. This emission comes from very small dust grains, with size < 0.01 microns. These grains are so small that the absorption of a single ultraviolet photon can produce a spike in their temperature to values much higher than would be expected for larger dust grains [\[21\]](#). There is also a broad absorption feature in the SEDs of all actively star forming galaxies at a wavelength of about 10 microns. This corresponds to a feature in the spectrum of silicates that are a major constituent of interstellar dust.

Almost all of the features we have discussed here can be seen in the SEDs of galaxies where star formation is still under way, be that in spiral galaxies like our own, or in galaxies called starbursts where star formation is taking place much more rapidly. We will learn more about starburst galaxies later in this chapter. Elliptical galaxies, however, usually do not host any star formation, and are largely lacking an interstellar medium. Because of this, their spectral energy distribution, from the mid-infrared to the far-infrared, is largely featureless. A small amount of dust, heated to fairly low temperatures of 20-30 K, can be found in them, and this produces a weak bump in their SED peaking at a wavelength of about 100 microns, but this represents an insignificant contribution to their energy output. Quite how elliptical galaxies come to have so little matter in their interstellar medium will be explored later in this chapter.

In star forming galaxies, the mid- and far-infrared can account for a significant fraction of the energy they emit. For a spiral galaxy, the mid- to far-infrared can be responsible for about 30% of the galaxy's energy emission. For the most active star forming galaxies, however, this fraction can be as high as 95%. The most actively star forming galaxies in the local universe may seem quite insignificant when observed in the optical, but can be very powerful emitters at longer, infrared wavelengths.

6.4 M31: OUR NEAREST NEIGHBOUR

The galaxy M31, found in the constellation Andromeda and therefore often referred to as the Andromeda galaxy, is the nearest large spiral to the Milky Way. It is likely that our own Galaxy would look pretty much like Andromeda if we were able to view it from the outside. The Andromeda galaxy lies about 2.5 million light years away, and can be seen as a fuzzy blob by the naked eye from a good site on a dark night. [Figure 6.2](#) shows what this galaxy looks like at a range of wavelengths, including the optical, far-infrared and X-rays. The light from stars can be seen in the optical, while the far-infrared shows the emission from dust, and X-rays show emission from the neutron stars and supernova remnants left behind at the end point of stellar evolution. If you compare the optical and far-infrared images, you will see that far-infrared emission is strongest where there are dark regions in the optical. These are dust lanes, where plentiful dust

absorbs much of the optical emission, reradiating it at longer far-infrared wavelengths. Gas is associated with this dust, so these same dust lanes are where star formation may be taking place.

By comparing the strength of the far-infrared emission in these dust lanes at different wavelengths, it is possible to measure the dust temperature [155]. This is shown in [Figure 6.3](#). As you can see, the centre, where there are the most stars to heat the dust, has the highest temperature at about 30 K. The rings of dust that coincide with the dust lanes are at lower temperatures, but there are blobs all along these dust lanes where the temperature rises to 20 K or more. Combining the Herschel dust images with observations at other wavelengths [61] shows that star formation is taking place throughout the dust lanes in Andromeda, but the places where the dust is warmest is where stars are forming most rapidly.

Most of this star formation is caused by the normal turbulent processes of the interstellar medium (see [Chapter 5](#)), but the detailed structure of the dust in the centre of this galaxy, combined with the ring of dust lanes at larger radii seen in [Figure 6.2](#), suggest this galaxy may have had a more violent past. These two rings taken together are thought to be the result of a collision between Andromeda and its smaller neighbouring galaxy M32 (which can be seen as the small fuzzy blob diagonally upwards and to the right of the big galaxy's centre in the optical image of Andromeda) which took place about 200 million years ago [18]. As we will see later, major collisions and mergers between galaxies can trigger large changes in their gas, dust and stellar content, as well as their shape. This more minor collision has left Andromeda largely unchanged, but M32 has lost much of its interstellar dust and gas. You can see this by comparing the optical and infrared images of M32. The smaller galaxy has far less dust, and thus far less gas, compared to its optical emission than its larger companion.

While this was just a minor incident, much larger things are in store for the Andromeda galaxy because our own Galaxy, the Milky Way, will collide with it in about four billion years [166]. This will be a much larger collision than Andromeda's encounter with M32. We will discuss what might happen as a result of this collision later in this chapter.

6.5 THE IRAS REVOLUTION

Studying our local neighbourhood in the universe is all very well, but we can only understand what is going on more broadly by observing galaxies across the whole sky. Before 1983, and the launch of the first infrared space mission, the far-infrared and much of the mid-infrared were unexplored territory. It was known that warm dust, detectable from the ground in the mid-infrared, was abundant in our own galaxy and was a good tracer of star formation, but very little was known about dust at longer wavelengths, and about its role in other galaxies. The Infrared Astronomical Satellite (IRAS) changed all that.

IRAS was the first satellite to observe the sky at mid- and far-infrared wavelengths. IRAS had only a 57 cm diameter main mirror, not much bigger than those in the largest amateur telescopes, but IRAS's mirror was not only in space, it was also cooled to liquid helium temperatures - just 4 degrees above absolute zero (i.e., -269 degrees Celsius), allowing it to make sensitive observations in the mid- and far-infrared. For a ten-month period following its launch on 25th January 1983 IRAS continuously scanned the sky, detecting sources at four different infrared wavelengths: 12, 25, 60 and 100 microns. While ground-based observations are possible at around 10 and 20 microns, IRAS was much more sensitive at these wavelengths, and was the first instrument to do significant work at 60 and 100 microns. The idea of making observations at these long wavelengths was a risk, since it wasn't clear what would be found, and getting the technology to work was a significant effort. The results, however, were spectacular. Many more sources were detected than were expected, and the scene was set for much of the work discussed in this book. IRAS discovered the first debris disk to be found, around the star Vega [10], found several comets [170] and uncovered the link between comets and meteor showers [160]. To my mind, though, it was IRAS's discovery of the 100 micron bump in the SEDs of spiral galaxies, and the even

more unexpected discovery of galaxies whose luminosity is dominated by the far-infrared, that was of most importance, since it set the scene for most of the future far-infrared space missions.

The majority of the IRAS satellite's 10-month operational life was spent on an all-sky survey, mapping out the sky in its four wavebands (see [Figure 6.4](#)). What the scientists behind the mission expected to find were dust clouds in our own galaxy, and to detect a few objects outside the Milky Way, such as the nearby Andromeda galaxy. They actually found many more faint point sources, located well away from the Galactic plane, than they were expecting. The first hint that these sources were outside our own galaxy came when the IRAS satellite scanned past the Large Magellanic cloud (LMC), a small satellite galaxy to the Milky Way. At this point the IRAS team did not have an accurate measurement of where the satellite's detectors were pointing [[145](#)], so their reaction on seeing the detector scans going crazy was that IRAS had detected a large cloud of debris in Earth orbit. Eventually they realised that they were in fact seeing several unexpectedly bright dust sources in the LMC, and were able to refine the pointing of the satellite - it turned out that an on-board clock was out by several seconds, leading them to think the satellite was pointing in a different direction. Once they had the correct positions, they were able to work out that the many point sources they were finding were usually galaxies already detected in optical surveys. Examination of the optical images revealed that most of these galaxies were spirals, and comparison of the optical and far-infrared data showed that roughly a third of the energy generated by stars in these sources was being absorbed by dust and reprocessed to longer, far-infrared, wavelengths.

6.6 THE NORMAL UNIVERSE

The picture presented in [Figure 6.5](#) is largely a product of the IRAS survey. Spiral galaxies form stars at a moderate rate of a few solar masses of new stars each year. The dust in the interstellar medium that is closest to the star formation regions absorbs more energy from the light of the new stars and heats up to temperatures of 30-40 K, while dust in other parts of the interstellar medium is heated to a lesser extent by the general light of stars in the galaxy, reaching a somewhat lower temperature of 20-30 K. This is what we see in our own Galaxy and in Andromeda, and the IRAS results show that more distant spiral galaxies behave in much the same way.

There are, however, a couple of things missing from this picture. Firstly, elliptical galaxies clearly don't work the same way, since they are largely lacking in interstellar medium, and do not have any significant ongoing star formation. Secondly, our discussion of spiral galaxies has so far been restricted to what is going on in their disks. If you look at the Hubble tuning fork diagram in [Figure 6.1](#) you will see that spiral galaxies have not only a flat disk of stars, in which you can find gas, dust and ongoing star formation, they also possess what is known as a bulge - a spheroidal-shaped distribution of stars distributed around the centre of the galaxy, and reaching greater distances above the galactic disk than stars from the disk itself. A key feature of the Hubble classification diagram is that spiral galaxies become more 'elliptical-like' as their bulges become a more significant part of their structure.

A good example of a galaxy with a very strong bulge and a very weak disk is the Sombrero Galaxy, shown in [Figure 6.4](#). This demonstrates a key aspect of disk galaxy bulges: Just like elliptical galaxies, bulges are made up almost entirely of stars, and contain very little gas and dust. In fact, in a number of ways, bulges look rather like little elliptical galaxies embedded within dusty, star-forming galaxy disks. This broadens the problem, but still leaves us with the core question: How and when did elliptical galaxies lose all their gas and make all their stars?

6.7 THE ACTIVE GALAXY MENAGERIE

Another thing that became clear from the IRAS all sky survey was that active galactic nuclei (AGNs), the sources at the centre of some galaxies powered by accretion onto a supermassive black hole, have surprisingly strong emission in the 12- and 25-micron bands. This indicates the presence of hot dust in these objects. More detailed analysis of the SEDs of a variety of AGN showed the presence of dust up to temperatures of 2000 K. This is surprisingly hot. In fact, it roughly matches the temperature at which dust should sublime - i.e., boil away directly from a solid into a gas. To get so hot, and to be emitting so strongly, the dust seen in an AGN would have to be very close, in astronomical terms, to the supermassive black hole at the core of an active galactic nucleus. The realisation that dust plays a significant role in the energetics of AGNs eventually solved one of the central puzzles that had been around since the original discovery of AGNs in the middle of the 20th Century: Why do AGNs have such a wide variety of properties?

The first AGNs discovered were spotted by Carl Seyfert [152] in the 1940s. He found a number of spiral galaxies with bright point-like emission in their nuclei, the spectra of which showed emission lines suggesting gas temperatures higher than could be produced in even the most luminous stars. More than that, some of the emission lines were also very broad, suggesting that the gas responsible for them was travelling at velocities in excess of several thousand km/s. Over the next several years Seyfert found more of these sources, now generally known as Seyfert galaxies, and he classified them into two types: Seyfert 1 galaxies with broad hydrogen emission lines as well as high excitation narrow lines, and Seyfert 2 galaxies with only the narrow high excitation lines (see [Figure 6.7](#)). As more data came in, this classification became more complicated, with some galaxies having weaker broad hydrogen lines than those classified as Seyfert 1s. These were classified as Seyfert 1.5 galaxies, and soon enough people were talking about even smaller gradations in classification, with Seyfert 1.8 and 1.9 galaxies.

While the optical astronomers studying relatively local galaxies were busy with the Seyferts, radio astronomy was getting going. The first large-area radio survey to be sensitive enough to detect a few hundred sources was the 3C survey (the third Cambridge radio survey), completed with accurate flux measurements in the early 1960s [15]. Gradually, over the next several years, optical counterparts to some of these radio sources were identified. Some of these counterparts were clearly galaxies when viewed in the optical - these included sources like Cygnus A, which is associated with an unusual elliptical galaxy that possesses a prominent dust lane at its centre. However, a number of other 3C radio sources were found to be associated with point-like optical objects, whose optical spectra included a large number of emission lines at wavelengths that defied identification. These sources were named 'quasi-stellar radio sources' (QSOs), a title often contracted to quasar.

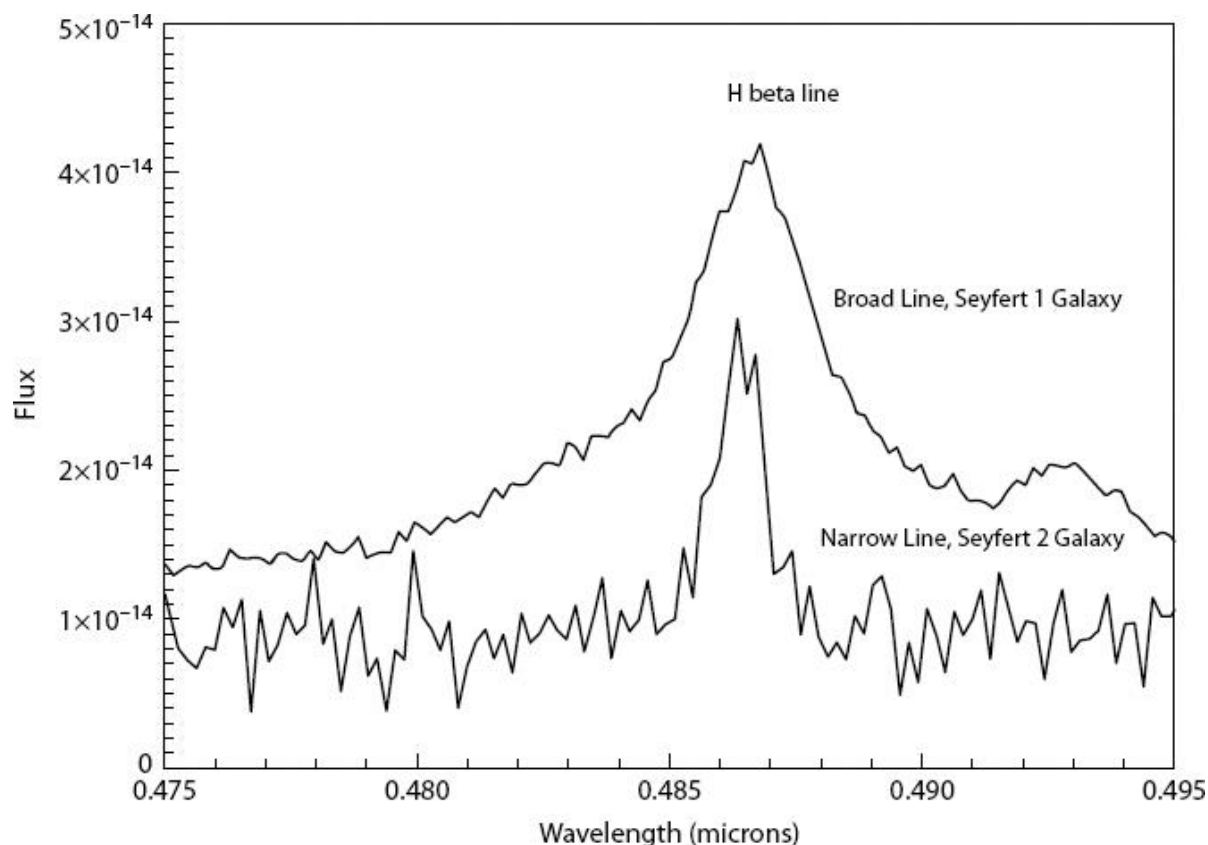


Figure 6.6 Example optical spectra of Seyfert galaxies.

These optical spectra show the distinction between broad line, Seyfert 1 type galaxies and narrow line Seyfert 2s. The line shown here is the H β line of hydrogen.

The mystery of QSO emission lines was finally solved when Maarten Schmidt realised that the emission lines in one of the brightest QSOs, called 3C273, could be identified if the lines were assumed to have been shifted red-wards, with the wavelength of each line multiplied by the same amount. It was already known that the spectra of galaxies are systematically shifted to longer wavelengths by an amount that directly depends on their distance from us. This is known as the redshift of a galaxy, and was discovered by Edwin Hubble in 1929. It is a consequence of the expansion of the universe, and provides a useful way of measuring the distance of galaxies. We shall discuss redshift more extensively in [Chapters 7](#) and [8](#). The redshift Schmidt found for 3C273 was 0.158 [[149](#)], an extremely high value at the time. When this high redshift was combined with the brightness of 3C273 it became clear that whatever was responsible for the emission in this source was very powerful indeed. We now know that quasars are powered by supermassive black holes, but it took at least two more decades before that idea was well established.

The detection of the first quasars through their radio emission was just a happy accident. Work over the decades subsequent to the discovery of 3C273 showed that only about 10% of quasars emit strongly at radio wavelengths - they are termed 'radio loud quasars' - and that the majority are in fact 'radio quiet', in the sense that they are not unusually powerful in the radio. The best way of finding quasars (and other types of AGN) is in fact to work at X-ray wavelengths, where they can be very prominent thanks to the high temperature emission of the material falling through their SMBH accretion disks.

By the end of the 1970s, astronomers studying AGN were faced with an ever increasing menagerie of different types of objects. Seyfert galaxies had been joined by radio loud and radio quiet quasars in the 1960s. Further studies of radio surveys like 3C had found a class of radio galaxies with prominent jets of radio emission, unexplainable by normal stellar processes². Some of these had broad emission lines in their optical spectra, and some had narrow lines. A few radio- or X-

ray-selected AGNs were found to vary very rapidly and to have featureless optical spectra, rising to shorter, bluer, wavelengths. One of these was sufficiently bright to have been listed as a blue, variable star in star catalogs of the 1920s, under the (incorrect) assumption that it lay in our own galaxy, and was given the name BL Lacertae (BL Lac). We now know that it lies in a galaxy at a redshift of 0.07 [124], and it has become the archetypal member of a class of AGN now known as BL Lac objects, or blazars.

This exploding zoo of different types of active galaxy was, as this brief and incomplete summary suggests, becoming unmanageable. The diversity of properties was so overwhelming that some unifying, underlying principle was needed to bring them back together. Without this, the idea that different types of AGN were all aspects of the same phenomenon might break down completely. Fortunately, detailed observations of the radio emission of blazars, arguably the most extreme AGN class of them all, provided some hints as to what that unifying principle might be. At first, though, these observations seemed to make matters worse, not better, and were initially interpreted as being a challenge to one of the central results of 20th century physics - the special theory of relativity, and the result that nothing can travel faster than the speed of light.

Observations of quasar radio jets at very high spatial resolution are able to track the movement of structures within the jets. By measuring how far these structures move along the jets over time, and taking the distance to the AGN derived from the redshift measured by optical spectroscopy, a physical velocity for these moving structures can be calculated. When this was done, a rather disturbing result was found: These blobs appeared to be moving faster than the speed of light (see Figure 6.7).

While some astronomers claimed that these apparently superluminal motions were evidence that the measured redshifts of quasars were not cosmological at all, it turns out that the answer to the apparent conflict with special relativity was already available. Some years before their observation, Martin Rees had shown that an observer looking straight down a column of material travelling towards them at close to the speed of light might perceive the material to be moving at superluminal speeds [140]. The superluminal motions weren't evidence against special relativity, they were in fact an optical illusion caused by it. The fact that these superluminal motions only occur when there is a very precise alignment between the direction of an emitted jet and the line of sight to the observer was consistent with the fact that they appear in only a very small fraction of AGNs. But what did this say about the broader population? Could it be that orientation effects, central to the perceived superluminal motions, might play a broader role in explaining the growing menagerie of different classes of AGN?

6.8 AGN UNIFICATION

An active galactic nucleus powered by the accretion of material onto a super-massive black hole can outshine an entire galaxy, but the central engine, where all the power is generated, is actually quite small. A region roughly the size of our own Solar System would encompass the black hole itself, the accretion disk and the region of rapidly rotating gas responsible for the broad emission lines. The narrow emission lines, however, lie much further away from this central engine, at least several light years, and in more luminous quasars potentially much further. This raises the possibility that something intermediate in size between that of the broad line region and that of the narrow line region might be able to obscure the emission from the accretion disk and the broad line region, but leave the narrow emission lines unaffected. The difference between a broad-line object, such as a quasar or a Seyfert 1, and a narrow-line object, such as a Seyfert 2, would be whether we were looking through this obscuring material or not. But you can't just make the emission from the central regions of the quasar disappear. These are compact objects that can, on their own, outshine the rest of the stars in their host galaxies. If the light from the central engine is going to be absorbed, it has to end up somewhere.

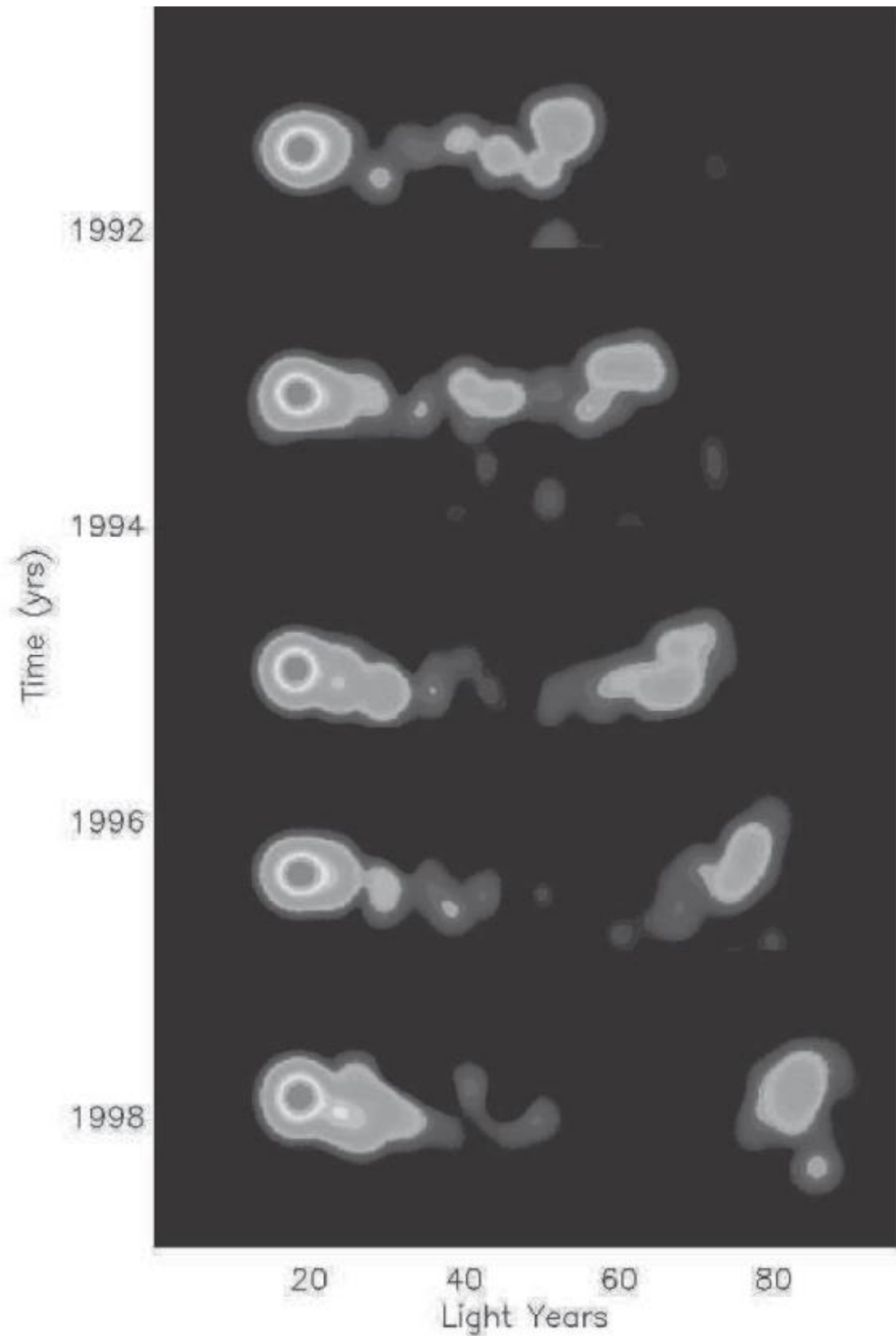


Figure 6.7 Radio images of the Blazar 3C279.

These radio images were taken over a period of six years, and appear to show structures, such as the blob at the end of this jet, moving at speeds greater than the speed of light. In this case the rightmost blob appears to move about 30 light years in

just six years time, suggesting that it is moving at five times the speed of light. This apparent contradiction with special relativity is in fact the result of an optical illusion, coming from the fact that we are looking right down a jet of material that is moving very close to, but not exceeding, the speed of light [175]. (Courtesy of NRAO; <https://public.nrao.edu/>.)

This is where the IRAS result, that AGNs contain a surprising amount of hot dust, proved significant. The hot dust seen by IRAS is being heated by the central engine of the AGN, and is indicating that dust, close to the central engine, is capable of absorbing a significant amount of its emission, reprocessing it from the X-rays, ultraviolet and optical, into longer wavelength, mid-infrared emission. If dust lay in a spherical distribution all around the central engine, then all AGN would be narrow line objects, as there would be no way for us to see through to the broad-line region or the accretion disk. Instead, the dust must lie in a structure like a doughnut - something known technically as a torus - so that some lines of sight can see the central engine and the broad emission lines, but others see only narrow line emission.

The idea that the orientation of our line of sight to the structures within an AGN determine how we see it, and how we classify it, has solved many of the issues surrounding the otherwise bewildering diversity of the AGN menagerie. The picture that has emerged of the structure of an AGN, and how we then classify what we see, is shown in [Figure 6.8](#). Viewing angles that intersect with the dust torus around the AGN see a narrow line object, those that can see past the torus to the central engine see a broad line object, while those most precisely aligned with the radio jet, in the 10% fraction of objects that emit such a jet, see superluminal motions and a BL Lac-type AGN.

This picture of the internal structure of an AGN also brings some interesting predictions, depending on the exact interaction between the dust in the obscuring torus and the emission of the central engine. If the dust acts like the interstellar dust we see in our own galaxy, then it should absorb light at shorter wavelengths more strongly than at longer wavelengths. This means that some AGN, aligned so that our view of their central engine only intersects with the edge of the obscuring dust, might reveal their central engine in the infrared, but hide it behind dust in the optical.

A good example of this is the powerful radio galaxy Centaurus A which, in the optical, appears as an elliptical galaxy with a prominent dust lane running across its centre. Radio jets are seen to emerge from the centre of the galaxy, presumably from an AGN, but optical light from that central engine cannot be seen because it is, presumably, absorbed by a dusty torus that surrounds it. Observations at longer wavelengths, though, can see through the obscuring dust and reveal a bright, compact source at the centre of the galaxy. This can be seen in [Figure 6.10](#). This has been done both from space, using the Spitzer Space Telescope [88], and from the ground using a mid-infrared camera on the 8m Gemini-S telescope [138]. The latter observations, capable of reaching a higher resolution thanks to the large size of the telescope, show that the bright nucleus of Centaurus A, which outshines the rest of the galaxy at mid-infrared wavelengths, must be less than about 10 light years across. Only an accreting supermassive black hole could produce such powerful emission.

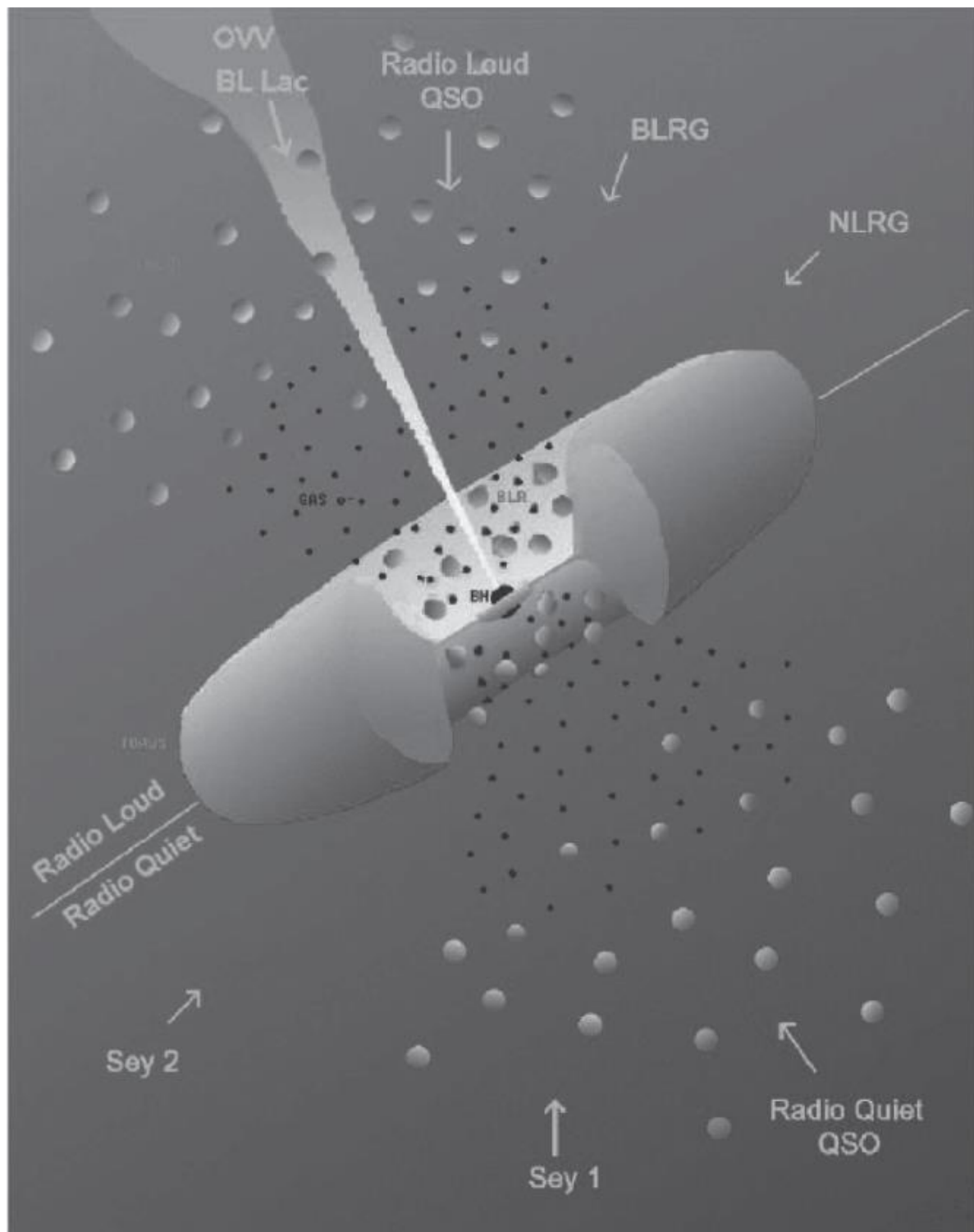


Figure 6.8 How the obscuring dust torus changes the observed appearance of an AGN.

(Copyright Astronomy Society of the Pacific, reprinted by kind permission of the authors [164].)

Another prediction of these so-called unified schemes is that some of the light from the central engine will be reflected by the obscuring torus rather than absorbed. This reflected light will not be very bright, but, because it has been reflected, it will be polarised, unlike the light received directly from the galaxy. For a few objects aligned in just the right way this means it might be possible to use reflected, polarised, light as a kind of periscope, able to peer into the depths of a central engine that could not be seen directly. This type of observation was first done in 1985 [5], when it was shown that the archetypical Seyfert 2 galaxy, NGC1068, had broad emission lines, the defining property of a Seyfert 1, when viewed in polarised light. A more modern example of such observations is shown in Figure 6.9.

What all these results show is that the various types of AGN are facets of the same phenomenon that appear different to us because we are viewing them from different angles. From some directions, when an AGN is being viewed through the obscuring torus of dust surrounding the central supermassive black hole, we cannot see any direct emission from the central engine. Instead, this light is absorbed by dust, which is heated to quite high temperatures, up to about 2000 K on the inner face of the dusty torus, a temperature so high that the dust is boiled away. Observations in the infrared, that can see this reprocessed emission, and using polarisation, that can, in some cases, see past the obscuration, have successfully unified the AGN menagerie.

6.9 GALAXIES AND AGN

While we only see AGN in a small fraction of galaxies, it is now generally accepted that nearly all galaxies contain a supermassive black hole at their centre, and that they all go through AGN phases during their lives. The supermassive black hole in the centre of a galaxy is ignited when a significant amount of material begins to fall onto it, triggering the formation of an accretion disk and driving all the other processes we have been discussing. As discussed above, our own galaxy has a supermassive black hole of about 4 million solar masses at its centre. If this were to ignite, it would be a fairly weak AGN - more powerful quasars are driven by supermassive black holes that are ten or more times as massive as that in the Milky Way.

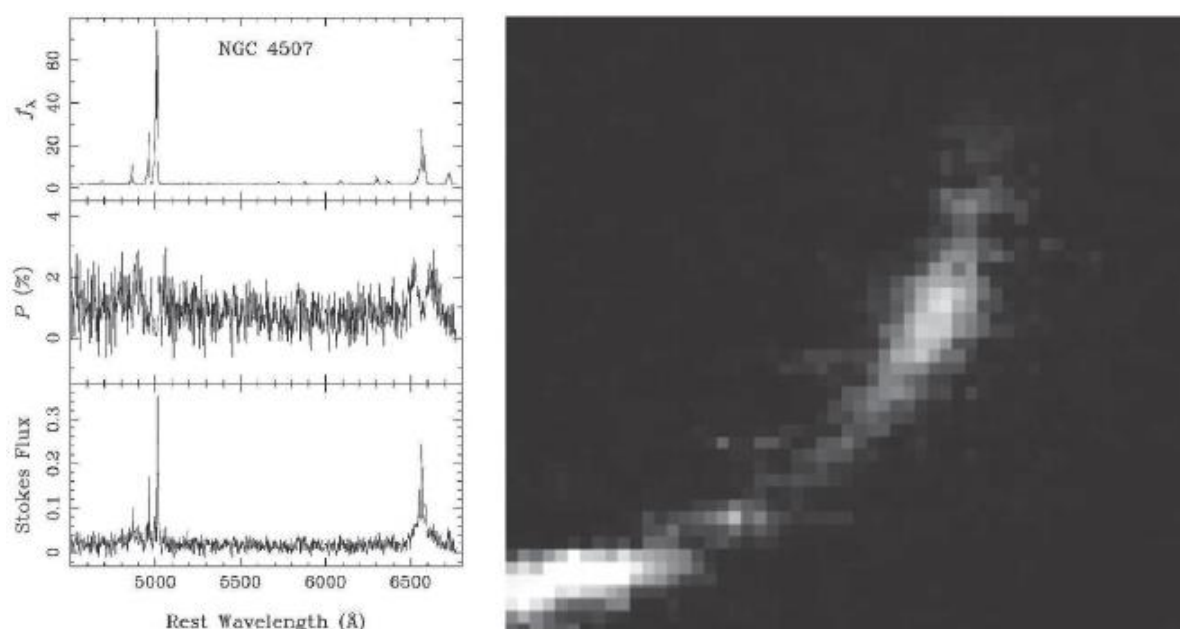


Figure 6.9 Left: The emergence of broad lines in polarised light. Right: Near infrared image of the gas cloud currently passing close to the black hole at the centre of our own galaxy.

Left: The top plot shows the unpolarised spectrum of the galaxy NGC4507. In this, the object appears as a normal Seyfert 2, with narrow emission lines. How the degree of polarisation changes with wavelength is shown in the middle figure, while the spectrum of polarised light is shown at the bottom. A much broader emission line emerges in this, which would classify the object as a Seyfert 1. The polarised spectrum is due to light being scattered off the dust torus from the otherwise obscured region around the central supermassive black hole. (Left: Reproduced by kind permission of Edward Moran. Right: Courtesy of ESO.)

There is, in fact, a cloud of gas currently (at the time of writing) passing very close to the Milky Way's supermassive black hole [65] (see [Figure 6.9](#)). At closest approach this cloud will be just 20 light hours away from the supermassive black hole. In Solar System terms, that is a distance of 144 AU, which puts this cloud of gas closer to the black hole than the Kuiper Belt object Sedna is to the Sun. While observations of this cloud clearly show that it is being ripped apart by the gravity of the black hole, there is, so far, no indication that any material from it has fallen onto the black hole. Even if it does, the total mass of gas in the cloud is small enough that the energy released will only amount to a short-lived burp in comparison to other AGN. But this does raise the more general question: what effects do AGN have on their host galaxy, and vice versa?

For a long time it was thought that the host galaxy could have an effect on the supermassive black hole - feeding it, or starving it for example - but there was little reaction in the other direction. After all, a galaxy is a large structure, many thousands of light years across, while even the most powerful AGN central engines are much less than a light year in size. Comparisons of the properties of galaxy bulges and supermassive black holes in local, non-active galaxies, have changed that view. A series of studies, starting with a publication in 1998 by Magorrian and collaborators [104], have found a clear relationship between the central black hole mass and the mass of stars in a galaxy's bulge. This suggests that the formation of the bulge of a galaxy, a structure that can be many thousands of light years in size, and the formation and growth of the galactic centre black hole are somehow inextricably linked. Fortunately, one of IRAS's other major discoveries provides some hints as to how this might occur.

6.10 STARBURST GALAXIES

The discovery of strong mid-infrared emission from AGN, and the subsequent development of unification schemes for the various classes of active galaxies, was just one of the major successes of IRAS. Another major surprise was the discovery of galaxies forming stars much, much more rapidly than is seen in quiescent spiral galaxies like the Milky Way. These objects are forming stars so quickly that they can form a Milky Way-sized galaxy, containing a hundred billion stars, in only a few billion years - much less time than the age of the universe. They will also exhaust the gas and dust supply of a typical spiral galaxy on a timescale of a few tens of millions of years. The star formation episodes we are seeing have to be of limited duration, so these sources are known as starburst galaxies.

M82, shown in [Figure 6.11](#), is the nearest starburst galaxy, at a distance of about 12 million light years from the Milky Way. It is a relatively small galaxy, with a total mass about one hundredth that of our own galaxy [69]. However, M82 is forming stars at a rate of 10 solar masses of new stars each year, ten times *faster* than the Milky Way. In fact, it is forming stars so rapidly that it seems to be blowing itself apart, with a strong wind of material flowing away from the central starburst regions of the galaxy. This outflowing material is revealed by emission from ionised hydrogen and is shown in red in [Figure 6.11](#). M82 turns out to be very bright in the far-infrared (see [Figure 6.5](#)), with roughly 90% of its total energy output emerging at these wavelengths. This is because much of the energy being emitted by the young stars born in M82's starburst is being absorbed by dust, and reradiated in the far-infrared.

M82, despite being a relatively minor galaxy in the grand scheme of things, provides several useful insights. Firstly, the wind of material being blown out by the starburst is an excellent example of how a high rate of star formation can provide feedback that acts to reduce the star formation rate, or even extinguish it completely. The starburst is blowing gas out of M82 that would otherwise lead to even more star formation, so the process is, in a sense, self-limiting. Secondly, when we try to find what might have triggered such an extreme star formation rate in such a small galaxy, we find evidence to suggest that M82 has had a close encounter with its larger neighbour, the galaxy M81. The two can be seen together in [Figure 6.11](#) in an image taken by the WISE satellite. In this image the colours red and green represent emission in the mid-

infrared, coming from warm dust, while blue represents light from stars. As you can see, despite being a much smaller galaxy than M81 - which is about 100 times more massive than M82 with roughly the same mass as the Milky Way - M82 outshines it in the mid-infrared, thanks to its prodigious star formation rate. The interaction between M81 and M82 will have caused gas in the centre of M82 to collapse, forming dense giant molecular clouds that are the site for the star formation driving its starburst. Without the interaction this gas would have remained in a stable orbit around M82's centre, forming stars at a much more sedate rate.

6.11 GALAXY MERGERS

M82 was a prominent IRAS source because it has a high star formation rate and because it is quite close to the Milky Way. Once detected by IRAS it was quite easy to identify its optical counterpart because it was already a well known object - the final Messier catalog, from which it takes its name, was published in 1781. The same could be said for many of the normal spiral galaxies that were detected by IRAS. However, many other IRAS point sources, some faint, but some very bright, were less easy to track down. When counterparts were sought by looking for optical sources at the positions of the IRAS objects, many were found to be faint anonymous blobs on the photographic plates available at the time. Others were found to have no optical identification at all. This led to a long-running series of campaigns to identify IRAS sources and to follow them up with optical imaging and spectroscopy.



Figure 6.10 The radio galaxy Centaurus A in optical (left) and mid-infrared (right).

(Left: Courtesy of NASA. Right: Courtesy of ESO.)



Figure 6.11 Left: M82, the nearest starburst galaxy. Right: M81 and M82 in the infrared.

Left: This optical and near-infrared image of M82 shows that it is a small disk galaxy, with plentiful dust in its disk. It also has a superwind of interstellar material blown out of the galactic disk, seen here as red emission tracing ionised hydrogen.

Right: An infrared image from the WISE satellite showing M82 (upper yellow object) and its larger neighbour M81. Blue shows the emission from starlight (3.4 and 4.6 microns), while green and yellow shows emission from the mid-infrared (12 and 22 microns). M82 is much brighter than M81 in the mid-infrared thanks to its very high star formation rate. (Courtesy of NASA.)



Figure 6.12 Arp220 (top) and the Antennae galaxy (bottom).

Top: An optical image showing the galaxy Arp220. Note the prominent dust lane across the centre and the unusual shape, which is unlike either a disk or an elliptical galaxy. (Courtesy of NASA.) Bottom: The Antennae galaxy is a merger between two spiral galaxies which has triggered a burst of star formation. (Courtesy of ESO.)

Much of this activity was kicked off when it was realised that one of the brightest extragalactic sources in the IRAS sky was associated with a galaxy known as Arp220. In the optical, this source isn't particularly bright. It has an unusual shape (see [Figure 6.12](#)), of which more later, which is why it was included in Arp's *Atlas of Peculiar Galaxies* [8], but at optical wavelengths it is otherwise undistinguished. In the far-infrared, though, Arp220 is an absolute monster, brighter than many galaxies much closer than its distance of roughly 250 million light years. M82 has a far-infrared luminosity equivalent to the power output of 30 billion suns; Arp220 is thirty times more luminous, emitting the power of a thousand billion suns in the far-infrared. In fact, as you can see from [Figure 6.5](#), the far-infrared is responsible for over 95% of the energy output of a major starburst like Arp220, which is why this galaxy looks so unexceptional in the optical, despite its prodigious energy output in the far-infrared. The power source for Arp220 is thought to be a massive starburst, as in M82, but in this case the star formation rate is a hundred solar masses of new stars each year - ten times the star formation rate of M82 and a hundred times the star formation rate of our own Galaxy. All of this star formation is buried beneath thick dense clouds of dust and gas, which is why all of the light they emit in the optical and ultraviolet is absorbed by dust and reradiated in the far-infrared. This has led to these sources being called 'dust dominated' since, although dust makes up only a small fraction of their mass, they are brightest at wavelengths where dust emits.

As more and more of the IRAS galaxies were identified, more and more galaxies like Arp220, called ultraluminous infrared galaxies (ULIRGs) because of their extreme far infrared luminosity (formally a ULIRG is a galaxy with far-infrared luminosity between 10^{12} , a thousand billion, and 10^{13} , ten thousand billion, times the luminosity of the Sun), were found. At the same time, more detailed followup observations of these ULIRGs were also undertaken, trying to find out what might have caused them to be so luminous. Hints from less luminous systems, like M82 or the Antennae galaxy (see [Figure 6.12](#)), which is a merger between two galaxies as well as a starburst about ten times as luminous as M82, suggested that galaxy interactions and mergers might trigger these massive starbursts. Confirmation that nearly every ULIRG found by IRAS was a galaxy interaction or merger took many years of followup, with the case not being fully proven until the mid-1990s (something I had a hand in myself [31]), more than a decade after the IRAS satellite had died.

More detailed studies of ULIRGs and other galaxy mergers, together with computer simulations of how stars, gas and dark matter react in a galaxy collision, have shown a number of things. Firstly, galaxy mergers will drive gas and dust into the centre of a galaxy, in much the same way that the weak interaction between M81 and M82 drove gas and dust into the centre of M82. This will promote a burst of star formation. The stronger the interaction, and the more dust and gas available to drive it, the bigger the starburst will be [12]. In principle, a major merger will be able to use up all the gas in the interstellar medium of a spiral galaxy. Secondly, the effect of the galaxy merger on the existing stellar populations in colliding disk galaxies is to destroy the rather delicate structure of the galactic disks. The end result of a violent merger between two gas-rich spiral galaxies turns out to look pretty much like an elliptical galaxy, in terms of both its shape and the lack of gas and dust in its interstellar medium. The question raised earlier in this chapter, about how elliptical galaxies came to have little or no interstellar medium, seems to have been answered.

But where do AGN come into this? Alongside the triggering of a major starburst, it seems likely that galaxy mergers are capable of driving gas deep into the core of a galaxy, fuelling the supermassive black hole lurking there and triggering AGN activity [146]. At the same time, the merger-induced starburst leads to the growth of the galaxy bulge with essentially the same structure as an elliptical galaxy. The simple coincidence of these two processes is not enough to produce the strong correlations between supermassive black hole and bulge masses seen in the nearby universe. For this to happen, the galactic centre AGN must take a more active role in terminating the star formation activity in its host galaxy. This is thought to occur in a process termed 'quasar feedback', whereby the energy generated by gravitational accretion onto the galactic centre black hole in an AGN drives powerful gas outflows into the host galaxy. These disrupt the giant molecular clouds formed around the galactic centre by the galaxy merger, ending the burst of star formation.

This process can probably be seen in action in the galaxy Markarian 231. This is a galaxy in the nearby universe, lying at a distance of about 600 million light years. Markarian 231 has pretty much everything - it is a broad line AGN, a ULIRG and, as you can see in [Figure 6.13](#), it is an interacting galaxy. What you can't see on these images, and which required far-infrared spectroscopic observations with the Herschel Space Observatory to find, is that there is a huge outflow of cool gas from the centre of this galaxy [57]. In this unusual object, we may be seeing all the processes that transform spiral galaxies into ellipticals underway at the same time: a galaxy merger, a massive burst of star formation, a new, powerful AGN lighting up at its core, and a massive outflow of material, driven by this AGN, that is even now beginning to turn off the starburst.

6.12 CONCLUSION

In this chapter we have looked at how observations across the infrared have revealed the processes that shape galaxies and the stellar populations within them, and have found how the IRAS satellite, which gave us the first all sky infrared survey, was instrumental in solving a wide range of problems behind the evolution of local galaxies. The importance of galaxy collisions and mergers was suspected before IRAS, but it took the observations of that satellite, and many years of followup studies of the sources it detected, to fully understand the linkages between galaxy mergers, star formation and AGN.

Studying these processes helps us understand how the universe came to be what we see today, but there is also a much more local reason why we might be interested in what happens during the merger of two large, gas rich spiral galaxies. That is because, in about four billion years, the Milky Way and the Andromeda Galaxy, M31, will collide and merge. That means our distant descendants, whatever they might be, will have a ringside seat to all of the processes we've been discussing here, with bursts of star formation, the ignition of galactic centre black holes as AGN, and the transformation of the two spiral galaxies into an elliptical. We're not going to be around to see it ourselves, but the view will be spectacular³.

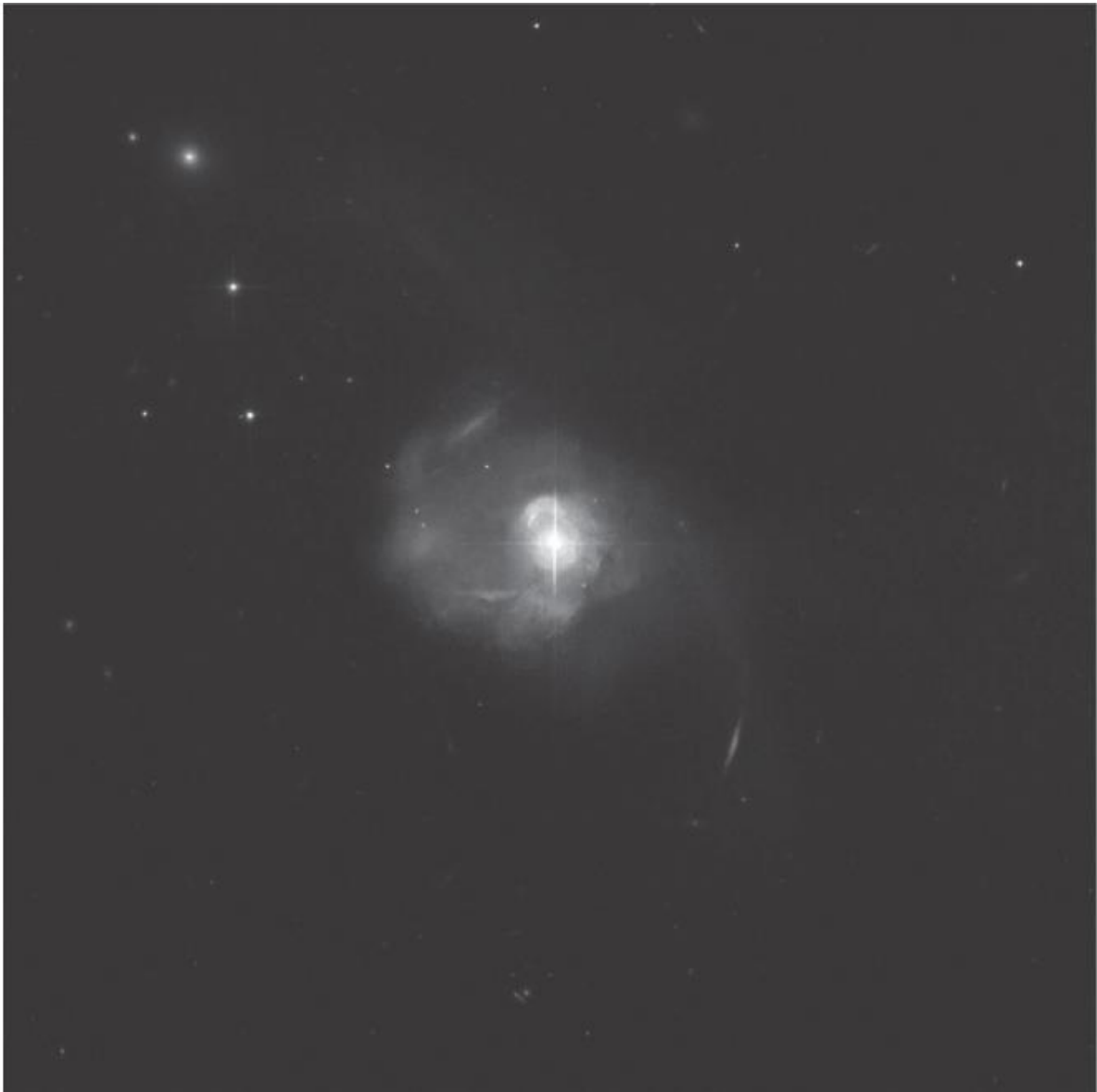


Figure 6.13 Markarian 231, the galaxy that has everything.

The bright, star-like source at the centre is the AGN, but this is surrounded by a galaxy merger hosting a massive burst of star formation. (Courtesy of NASA.)

¹ Fullerenes were first discovered in a lab experiment trying to recreate spectroscopic features seen in the interstellar medium with laser-blasted graphite. While these molecules didn't explain all the features that we now know are due to PAHs, they did revolutionise carbon chemistry and start new industries aiming to exploit fullerenes for practical applications.

² In fact there are two classes of radio galaxies, known as FR1 and FR2, as defined by the spatial structure of their radio emission, but that isn't relevant to the current discussion.

³ A preview of what this interaction might look like has been produced by some of the people who simulate galaxy mergers, and is available via a link on this book's webpages.